

2.3 SLICE THICKNESS, GAP, NUMBER OF SLICES & COVERAGE

HOW MUCH ANATOMY DO YOU IMAGE — AND HOW WELL?

Slice thickness, gap (spacing), and number of slices determine 3D coverage, partial volume effects, scan time, and SNR. The technologist balances coverage needs with image quality and scan efficiency.

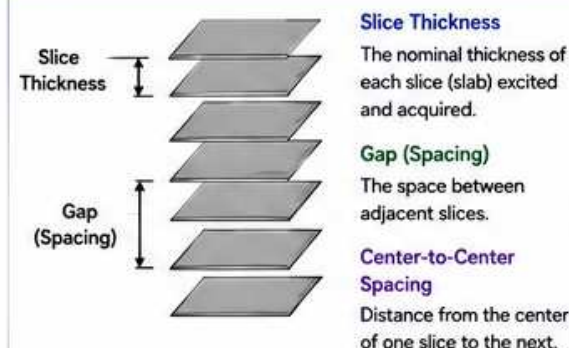


KEY CONCEPT

Thinner slices = more detail but lower SNR and longer scan.
Gap reduces cross-talk but can miss anatomy.
More slices = more coverage but more time.

↑ Increase
↓ Decrease
— No / Minimal Effect

1 DEFINITIONS



$$\text{Center-to-Center Spacing} = \text{Slice Thickness} + \text{Gap}$$

2 EFFECTS OF CHANGING SLICE THICKNESS

Change	Effect on SNR	Effect on Detail (Resolution)	Partial Volume Effect	Scan Time
Thinner Slices	↓	↑	↓	↑
Better lesion conspicuity, more anatomic detail, but lower SNR and longer scan.				
Thicker Slices	↑	↓	↑	↓
Higher SNR and shorter scan, but more partial volume and less detail.				

3 GAP (SPACING) CONSIDERATIONS

With Gap	No Gap (Contiguous)
- Reduces cross-talk between slices - Less slice overlap/blurring - Total coverage is slightly less	- Maximum coverage - Slightly more cross-talk (usually minimal) - Preferred for thin slices
 Coverage <	 Coverage =

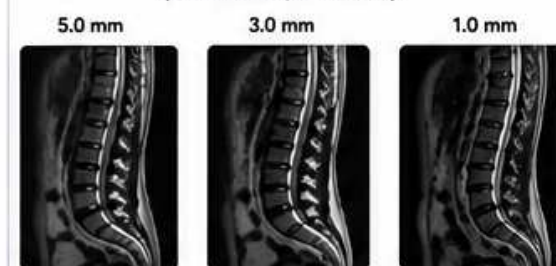
4 MORE SLICES = MORE COVERAGE

- Number of slices determines how far the exam extends.
- More slices → more anatomy → longer scan time.
- Coverage must include area of clinical interest with margins.



5 VISUAL EXAMPLES

A. SLICE THICKNESS (FOV 220 mm, Matrix 256)

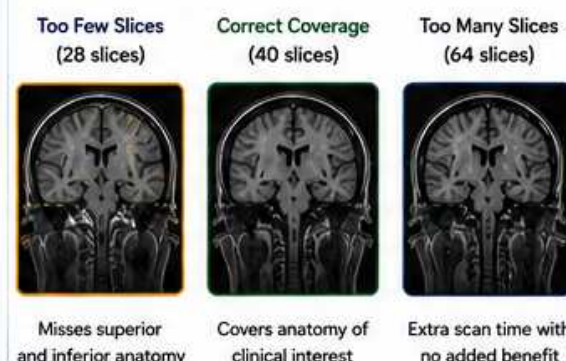


Thicker = less detail Thinner = more detail

B. GAP (WITH SAME SLICE THICKNESS 3.0 mm)



C NUMBER OF SLICES (SAME PARAMETERS)



D PARTIAL VOLUME EFFECT



6 TRADE-OFF SUMMARY

Parameter	Increase	Decrease	Effect On...
Slice Thickness (thinner)	↓ SNR	↑ Detail	Scan Time ↑ Partial Volume ↓
Gap (larger)	↓ Cross-talk ↓ Slice Overlap	↑ Coverage	Minimal increase in Scan Time
Number of Slices (more)	↑ Coverage	↓ Scan Time Efficiency	Scan Time ↑

7 COVERAGE CALCULATION

$$\text{Total Coverage (approx.)} = \text{Number of Slices} \times \text{Center-to-Center Spacing}$$

Example:
 • 40 slices
 • Slice Thickness = 3.0 mm
 • Gap = 1.0 mm
 Center-to-Center Spacing = 3.0 + 1.0 = 4.0 mm
 Total Coverage ≈ 40 × 4.0 = 160 mm
 (Allow margin beyond anatomy of interest)

8 PRACTICAL GUIDELINES

- Use the thinnest slice that provides adequate SNR.
For routine brain: 3–5 mm | Spine: 3 mm | Joints: 2–3 mm | 3D: 1–2 mm
- Use a small gap (0–20% of slice thickness) if cross-talk is a concern.
- Ensure coverage includes anatomy of interest with margin.
- Avoid more slices than needed — it only increases scan time.

9 ENGINEER CHECKLIST

- ✓ Is slice thickness appropriate for the exam?
- ✓ Is gap appropriate (too large = missing, too small = cross-talk)?
- ✓ Is coverage complete in all 3 dimensions?
- ✓ Is the number of slices optimized for the task?
- ✓ Are scan time and SNR acceptable?



KEY TAKEAWAY

Slice thickness, gap, and number of slices work together to control coverage, image quality, and scan time. Understand the trade-offs to make smart protocol choices.